

VEIHOLMEN, Norway

WMO: 012250

Lat: 63.517N

Lon: 7.950E

Elev: 16

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.4	26.5	12.4	10.3	26.1	15.2	11.9	28.3	46.8	41.8	42.4	41.6	14.1	60	0.821

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	66.4	58.9	63.7	57.7	61.8	57.0	60.2	64.5	58.9	62.7	57.7	61.1	10.2	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.2	72.5	62.4	56.9	69.2	61.0	55.7	66.1	59.9	26.5	64.9	25.7	62.7	24.9	61.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 43.6	39.0	34.8	22.2	73.2	4.8	3.5	18.7	75.8	15.9	77.8	13.2	79.8	9.7	82.3		
(5)			19.3	62.0	4.2	1.9	16.2	63.4	13.7	64.5	11.4	65.5	8.3	66.9	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	37.7	37.4	38.2	42.3	46.9	51.5	56.2	57.4	54.5	48.4	42.8	39.7
	DBStd	8.34	4.63	5.00	4.47	4.11	4.30	3.51	3.88	3.18	3.48	4.37	5.17	5.30
	HDD50	2092	382	354	365	234	115	20	0	0	5	80	218	319
	HDD65	6887	847	774	830	680	560	404	276	236	316	513	667	784
	CDD50	684	0	0	0	4	20	66	191	230	139	32	2	0
	CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDH74	3	0	0	0	0	0	0	3	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	15.5	17.5	16.6	17.4	14.2	13.0	13.6	12.4	12.3	15.7	16.4	17.4	19.1
Precipitation	PrecAvg	64.4	6.8	5.9	5.7	3.5	3.1	4.1	4.2	5.4	6.0	6.0	6.0	7.7
	PrecMax	77.4	16.6	11.7	12.4	10.3	5.8	7.2	7.0	10.8	15.4	11.0	15.0	15.0
	PrecMin	43.5	0.2	0.4	1.9	0.9	1.3	1.4	1.7	1.7	2.2	1.3	0.4	2.0
	PrecStd	8.1	3.7	2.8	2.8	2.1	1.1	1.6	1.4	2.1	3.2	2.2	3.9	3.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.7	47.0	48.8	58.5	63.5	65.3	72.1	70.0	64.8	61.9	53.5	50.3
		MCWB	43.3	42.5	42.2	48.9	53.3	55.7	61.4	59.7	56.4	52.6	49.7	46.1
	2%	DB	46.2	45.7	46.4	53.4	58.0	60.7	67.2	66.0	62.1	57.5	52.1	48.8
		MCWB	42.0	42.3	42.2	46.6	50.9	54.8	60.1	59.2	55.8	52.3	48.2	45.5
	5%	DB	45.2	44.6	45.3	49.6	54.7	58.0	63.6	63.6	60.4	55.6	50.6	47.8
		MCWB	41.6	41.4	41.8	44.4	49.4	53.4	58.5	58.3	55.4	51.2	46.9	44.4
	10%	DB	44.1	43.4	44.2	47.0	52.3	55.7	61.4	61.8	59.0	54.1	49.3	46.5
		MCWB	40.8	40.2	41.1	43.2	48.1	52.2	57.6	57.6	55.0	50.0	45.7	42.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.9	44.7	44.9	49.0	54.0	57.6	63.4	61.8	58.5	54.6	50.9	47.7
		MCDB	46.7	45.8	46.2	55.3	61.0	63.5	70.1	66.2	61.8	57.9	52.7	49.2
	2%	WB	43.6	43.2	43.8	46.6	51.6	55.3	60.5	60.2	57.3	52.9	49.2	46.4
		MCDB	45.6	44.6	45.3	51.4	56.8	59.3	65.5	64.1	60.0	56.6	51.4	48.4
	5%	WB	42.3	41.6	42.6	44.9	50.0	53.8	59.2	59.0	56.4	51.8	47.5	44.9
		MCDB	44.5	43.6	44.6	47.8	54.2	57.0	63.1	62.7	59.3	55.6	50.2	47.6
	10%	WB	41.2	40.4	41.3	43.8	48.5	52.5	57.9	57.9	55.5	50.4	45.9	43.2
		MCDB	43.8	42.7	43.9	46.3	51.5	55.4	61.5	61.4	58.6	53.9	48.9	46.0
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.4	5.8	5.7	5.6	4.8	5.0	5.0	5.3	5.2	5.2	5.6
		MCDBR	5.6	5.8	6.6	10.3	10.6	9.4	10.4	8.8	7.4	6.7	5.4	6.5
		MCWBR	5.3	5.4	5.1	5.6	5.5	5.1	5.1	4.5	4.5	4.8	5.1	6.2
	5% WB	MCDBR	5.4	5.3	5.8	8.0	9.7	7.9	9.3	7.5	6.3	6.2	5.4	6.2
		MCWBR	5.3	5.6	5.3	5.1	5.4	4.8	5.0	4.3	4.4	4.9	5.5	6.3
Clear-Sky Solar Irradiance	taub	0.397	0.316	0.339	0.359	0.354	0.355	0.364	0.355	0.341	0.315	0.368	0.358	
	taud	2.661	2.262	2.338	2.379	2.456	2.485	2.487	2.512	2.520	2.411	2.596	2.469	
	Ebn at Noon	107	193	239	260	273	275	269	261	241	195	105	51	
	Edh at Noon	10	19	25	30	30	30	30	26	22	16	9	5	
All-Sky Solar Radiation	RadAvg	41	188	564	1095	1508	1618	1481	1135	671	291	69	20	
	RadStd	6	33	84	98	68	161	133	108	78	43	13	2	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (30 neighbors)	+0.77	+1.21	N/A	N/A	N/A	N/A	-250	-329	+104	N/A

Nomenclature: See separate page